

Atiqur Rahaman

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Education

Bangladesh University of Engineering & Technology (BUET)
B.Sc. in Naval Architecture and Marine Engineering
CGPA: 3.78/4.00 (3rd in the Department)

Bangladesh
Jan 2022 – Jun 2026

Research Interests

CFD • Phase Changing Material • Heat Transfer • Battery Thermal Management • Offshore Structure

Publications

1. Md Tanvir Anjum Zenith, Adri Das, Shafollo Proshad Gupto, **Md. Atiqur Rahaman**, Md. Khairul Islam, “Numerical Investigation of Melting Enhancement in a Phase Change Material-Filled Enclosure Using K-Shaped Fins: Effects of Fin Angle and Arm Length Ratio” (*Under Review*). *Case Studies in Thermal Engineering*.
 - Studied K-shaped fins in a PCM enclosure in **ANSYS Fluent 2024R1** across a range of fin angles and arm length ratios; intermediate angles with a longer lower arm raised melting efficiency.
 - Contribution: conceptualization, ran the simulations, formal analysis and data curation, and produced the contour plots, efficiency comparisons, and LMLI curves.
2. Dr. Md. Shahidul Islam, **Md. Atiqur Rahaman**, and Md Tanvir Anjum Zenith, “Effect of Mooring Line Angular Spacing on the Wave-Frequency Motion Response of a 15 MW Semi-Submersible Floating Offshore Wind Turbine Based on AQWA” (*In Preparation*).
 - Examined how mooring spread angle affects the motion response of a semi-submersible platform in **ANSYS AQWA**; tighter spreads cut surge and yaw in head seas, with the effect fading or reversing at other headings.
 - Contribution: co-developed the methodology, built and validated the model, ran the simulation matrix, handled the analysis and visualization, and contributed to writing and project administration.

Research Experiences

Undergraduate Thesis Student, NAME, BUET

Apr 2025 – Jun 2026

Supervisor: Dr. Md. Shahidul Islam

Numerical Investigation of Mooring Line Angular Spacing on the Dynamic Motion of a 15MW Semi-Submersible Offshore Wind Turbine Platform

A hydrodynamic study of edge turbines, which face undisturbed waves and reach shutdown limits earlier than sheltered interior turbines.

- Built a coupled frequency domain and time domain model of the IEA 15 MW reference turbine on the UMaine VoltturnUS-S platform in **ANSYS AQWA**, validated against published data, and found surge and yaw to be the modes most sensitive to spread angle while heave, roll, and pitch stayed largely unaffected.

Ongoing Research

Aug 2026 – Present

Iso-Volume Comparison of Fin Geometries for PCM-Based Battery Thermal Management: Shape Ranking under Repeated Charge and Discharge Cycling

- This study uses **ANSYS Fluent** to measure how long each battery, PCM, and fin arrangement can hold the cell below its safe operating temperature. Every arrangement is given the same amount of metal, so the comparison will reflect the fin layout itself and not how much material each design uses.
- The comparison runs over many charge and discharge cycles rather than a single one, tracking how much of the PCM stays melted between cycles, since an arrangement that holds the temperature well at first may lose that ability once the PCM no longer fully recovers.

Projects

Undergraduate Project

Jan 2024 – Feb 2025

Supervisor: Dr. Mir Tareque Ali

Design of a 175 Passenger Ferry for circular waterways around Dhaka

- Carried out a full concept to preliminary design of a passenger ferry, working through the design spiral from principal dimensions and lines plan to hydrostatics, resistance, powering, weight, and stability in linked Excel sheets, and designed the propulsion system covering propeller selection with a cavitation check and sizing of the engine, gearbox, and foundation.

Work Experience

Intern

Jan 2024 – Feb 2025

Dockyard & Engineering Works Ltd, BANGLADESH NAVY, Narayanganj, Bangladesh

- Completed a three week industrial attachment at a working shipyard, with training on marine propulsion systems, shipboard air conditioning, thermal management of engine rooms, and CNC plate cutting.

Honors and Awards

- University Merit Scholarship (8 times in 8 semesters)
- University Dean's List Scholarship (2 times in 1st and 2nd year)
- Education Board Technical Scholarship, BUET.
- 1st prize in "NAME OLYMPIAD" at NAME FEST 2023
- 1st prize in "Worldwide Ferry Safety Association Design Competition 2025", Team Black Pearl.
- 3rd Prize in "Worldwide Ferry Safety Association Design Competition 2026", Team Black Pearl.
- Received award and certificate from the Chairman of BIWTC in recognition of outstanding performance in an international competition (Ferry Safety Association Design Competition).

Extracurricular Activities

- University Class Representative, Department of Naval Architecture and Marine Engineering, BUET Jan 2022 – Feb 2023
- Volunteer, MARTEC 2022 – International Conference on Marine Technology Dec 2022
- Team Member, Black Pearl, BUET, Worldwide Ferry Safety Association Design Competition Sept 2024 – Present
- Former Vice President, BUET Photographic Society Jun 2025 – July 2026
- Former Director of Design and Creation, BUET Entrepreneurship Development Club Jun 2025 – July 2026

Skills

Simulation Software	ANSYS FLUENT, ANSYS AQWA, ABAQUS
Programming languages	C++, Python, Fortran.
CAD & 3D Modelling	AutoCAD, Rhinoceros 3D
Ship design-related Software	MAXSURF, DELFTship.
Referencing Software	Zotero
Soft Skills	Leadership, Team Work, Mass Communication

References

1. Dr. Mir Tareque Ali
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2. Dr. Md. Shahidul Islam
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